COURSE TITLE : TECHNOLOGY OF ELASTOMERS
COURSE CODE : 3094
COURSE CATEGORY : B
PERIODS/WEEK : 6
PERIODS/SEMESTER: 90/3
CREDITS : 5

TIME SCHEDULE

Module	Topics	Period
1	Compounding ingredients	25
2	Rubber compounding	31
3	Cure characteristics and vulcanisation methods	17
4	Compounding for the vulcanizate properties	17
TOTAL		90

COURSE OUTCOME

Sl.	GO	Student will be able to
1	1	To Comprehend the preparation , properties and applications of compounding ingredients
2	2	To understand the principles of rubber compounding and vulcanisation
3	3	To understand the cure characteristics ,cure measurements and different vulcanisation methods
4	4	Able to apply principles of compounding for the design of various recipes to meet the vulcanizate properties

SPECIFIC OUTCOME

MODULE I COMPOUNDING INGREDIENTS

1.1.0 To Comprehend the preparation , properties and applications of compounding ingredients
- 1.1.1 Classify the compounding ingredients.
- 1.1.2 State the functions and characteristics of compounding ingredients with examples.
- 1.1.3 Explain different vulcanising agents and their roles in a recipe.
- 1.1.4 Describe the functions of activators and retarders.
- 1.1.5 State the function of accelerators and their classifications.
- 1.1.6 Explain the classification and role of antidegradents, processing aid, (plasticiser, softener, and extender).
- 1.1.7 Classify fillers with respect to colour, reinforcement, origin

1.1.8 Describe the preparation, characteristics and application of nonblack fillers like inorganic, fibrous, organic. resinous fillers.
1.1.9 Explain the classification and method of manufacturing carbon blacks.
1.1.10 State the properties of carbon black, and classification according to ASTM D 1765.
1.1.11 Describe the procedure for the determination of particle size and structure.
1.1.12 List the special purpose additives and their functions.

MODULE II RUBBER COMPOUNDING

2.1.0 To understand the principles of rubber compounding and vulcanisation

2.1.1. Define compounding and its objectives.
2.1.2. Explain the various steps involved in the design of recipes for product manufacture.
2.1.3. State the importance of base polymer in a compound design.
2.1.4. Explain mastication, masterbatch, gum compound and filled compound.
2.1.5. Justify the curing of elastomers.
2.1.6. Explain the type of crosslinks formed and their effect on vulcanisate properties for sulphur and nonsulphur curing systems.
2.1.7. Distinguish CV, EV, and Semi-EV curing systems.
2.1.8. Explain the chemistry of sulphur vulcanisation.
2.1.9. Select suitable curing systems for Natural and synthetic rubbers based on sulphur and nonsulphur systems.

MODULE III CURE CHARACTERISTICS AND VULCANISATION METHODS.

3.1.0. To understand the cure characteristics ,cure measurements and different vulcanisation methods

3.1.1. Define scorch and scorch time, cure and cure time, optimum cure, state of cure, cure index, delayed action.
3.1.2. State the methods for avoiding scorch.
3.1.3. Illustrate cure graph of mooney viscometer and rheometer.
3.1.4. Explain the various cure graphs by changing different type of accelerators and curing agents.
3.1.5. Describe the procedure for the determination of cure characteristics.
3.1.6. Define viscosity, elasticity and plasticity.
3.1.7. Describe different basic processing steps for product manufacture using NR and SR.
3.1.8. Explain different heating medium, mention their merits and demerits.
3.1.9. Define moulding, and state compression moulding, transfer moulding and injection moulding.
3.1.10. State vulcanisation methods other than moulding.
3.1.11. Explain different batch curing methods.
3.1.12. Describe continuous vulcanisation methods such as fluidised bed, microwave,rotocure, steam tube curing, molten salt bath with suitable examples.

MODULE IV COMPOUNDING FOR THE VULCANIZATE PROPERTIES

4.1.0. Able to apply principles of compounding for the design of various recipes to meet the vulcanizate properties
4.1.1. Explain the general principles of compounding, chart out the properties of elastomers.
4.1.2. Describe the elements to be considered for preparing a formulation and a product.
4.1.3. Describe the method for calculation of specific gravity and volume cost of a compound with examples.
4.1.4. Explain compounding to meet processing requirements.
4.1.5. Explain the effect of particle size and structure of filler on processing and vulcanisate properties.
4.1.6. Design compounds for a particular 'shore A' hardness and modulus using black and nonblack fillers.
4.1.7. Describe the factors to be considered for compounding specific vulcanisate properties such as abrasion resistance, hardness, tensile properties, tear resistance, oil resistance, heat,light,flame, oxygen and ozone resistance.

CONTENT DETAILS

MODULE-I

COMPOUNDING INGREDIENTS

Compounding ingredients- classification- base polymer, activators and retarders, accelerators, filler, processing aid, anti degradents, vulcanizing agents, and special additives-examples, characteristics and their functions in a compound. Selection procedure of ingredients- compatibility and solubility parameters.
Vulcanising agents-sulphur, sulphur donor, peroxide, metal oxide, resin, radiation. Carbon black- types-furnace, thermal, channel and lamp black- method of manufacture-furnace process and thermal process. Properties- particle size, structure, physical nature of surface, chemical nature of surface and particle porosity. Classification- ASTM D 1765. Procedure- particle size, structure.

MODULE-II

RUBBER COMPOUNDING

Compounding- definition- objectives. Compound design- recipe for products- examples. Mastication, master batch, gum and filled compound. Curing of elastomers- types of crosslinks- sulpher curing systems- CV, EV, and Semi EV, Chemistry of sulphur curing, Non-sulphur curing system- CR, CSM, EPM, vulcanisate properties.

MODULE-III

CURING CHARACTERISTICS AND VULCANIZATION METHODS

Curing- scorch, scorch time, cure and cure time, optimum cure, state of cure, cure index, delayed action. Scorch safety- methods. Cure graph- Mooney viscometer, rheometer, cure curves- by changing accelerators and curing agents. Definitions- viscosity, elasticity and plasticity. Product manufacture- basic steps. Different heating medium, merits and demerits- steam, electrical and thermic fluid. Moulding- compression, transfer and injection. Batch curing and continuous curing methods- examples.

MODULE -IV

COMPOUNDING FOR THE VULCANIZATE PROPERTIES

General principle of compounding – chart out properties of elastomers . products-formulation, elements to be considered for preparing a formulation . specific gravity and volume cost-calculation, examples . Compounding to meet processing and vulcanizate properties- hardness, tensile, tear resistance , oil , heat, light, flame .oxygen and ozone resistance. Compound design –for hardness, modulus –black and non black fillers.

REFERENCE BOOKS

1. Rubber Technology And Manufacture- C.M.Blow (Publisher – Butterworth Hainemann Ltd, 1982 Edition)
2. Rubber Technology -Maurice Morton (Publisher – Springer, 1999 Edition)
3. Rubber Technology Hand book - Werner Hoffman (Publisher – Oxford University Press,1989 Edition)
4. Vanderbilt Rubber Hand book -R.T Vanderbilt (Publisher – R.T Vanderbilt, 1990 Edition)